

Program : Diploma in Textile Technology	
Course Code : 6069	Course Title: Fabric Manufacture III Practical
Semester : 6	Credits: 1.5
Course Category: Program Core	
Periods per week: 3 (L:0, T:0, P:3)	Periods per semester: 45

Course Objectives:

- To provide thorough knowledge on analyzing the jacquard sample cloths and preparation of jacquard for weaving.
- Impart knowledge for setting the various motions in multiple box, Automatic and Rapier loom.
- Analyzing and costing of fabric and operation of CAD.

Course Prerequisites:

Topic	Course code	Course name	Semester
Knowledge of basic Mathematics		Mathematics 1	1
Computing fundamentals		Introduction to IT system	1
Knowledge on jacquard looms		Fabric manufacture II	4

Course Outcomes:

On completion of the course, the student will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Analyzing the jacquard sample cloths and preparation of jacquard for weaving.	11	Applying
CO2	Demonstrate the setting of multiple box, Automatic and Rapier loom.	11	Applying
CO3	Analyze and estimate the costs of different fabrics	10	Applying
CO4	Familiarize with the operation of CAD in fabric design.	11	Applying
	Lab Exam	2	

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	3					
CO2	3	3		2			
CO3	3	3					
CO4	3	3		3			

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

On completion of the course, the student will be able to:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Analyzing the jacquard sample cloths and preparation of jacquard for weaving.		
M1.01	Develop a full Jacquard design from a motif figure and prepare the card and perform the card lacing to weave the design.	3	Applying
M1.02	Study the single lift single cylinder Jacquard and Identify various parts of single lift single cylinder Jacquard	3	Applying
M1.03	Study the double lift double cylinder Jacquard and Identify various parts of double lift double cylinder Jacquard	3	Applying
M1.04	Analyze different Jacquard sample and prepare the details required to reproduce the same (minimum 10 samples)	2	Applying
CO2	Demonstrate the setting of multiple box, Automatic and Rapier loom.		
M2.01	Practice the setting of various operations in an automatic loom and operate the loom	3	Applying
M2.02	Prepare the pattern card for drop box loom for various weft colouring schemes and practice the setting of drop box as per timing.	3	Applying
M2.03	Familiarize with the working of Rapier picking mechanism	3	Applying
M2.04	Practice the setting of single thread warp stop motion Rapier picking mechanism	2	Applying
	Lab Examt -1	1	

CO3	Analyze and estimate the costs of different fabrics.		
M3.01	Study the factors influencing costing and Estimate the Ex-mill cost one meter of grey fabric	3	Applying
M3.02	Estimate the Ex-mill cost one meter of bleached and dyed fabric	3	Applying
M3.03	Estimate the Ex-mill cost one meter of check fabric	2	Applying
M3.04	Estimate the cost of production by analyzing the given sample cloths	2	Applying
CO4	Familiarize with the operation of CAD in fabric design.		
M4.01	Familiarize with computer aided woven design software, colour library and yarn library	3	Applying
M4.02	Use the software for dobby design	3	Applying
M4.03	Use the software for Jacquard design	3	Applying
M4.04	Familiarize with Fabric simulation	2	Applying
	Lab Exam - II	1	

The students should keep separate record for Fabric manufacture and submit certified & bonafide record at the time of practical examination

Text / Reference:

T/R	Author/Book Title
R1	Thomas W. Fox - The Mechanism of Weaving - Universal Publishing corp.
R2	NCUTE - Woven Fabric Production II - NCUTE New Delhi
R3	R.Sengupta - Weaving Calculations - D.B.Taraporevala sons & co Ltd.,
R4	Prof. J.C.Chakravorthy - Mechanism of Weaving Vol- I&II - Prof. J.C.Chakravorthy West Bengal
R5	Julie Holyoke's - Digital Jacquard Design-surface design association

Online Resources:

Sl. No	Website Link
1	http://www.nptel.ac.in/courses
2	http://freetexbooks.blogspot.in/2011/10/handbook-of-weaving.html
3	http://www.lynda.com/software/all
4	http://textilelearner.blogspot.in
5	https://www.textronic.com